CODENEXUS / TEAM CHROLLO / NSEC

Dante

Brand change and drift tracking

01 Understanding of the problem statement

Brand facts, rules, terms, and approved assets are scattered across documents and teams. A reviewer may catch a wrong claim in a new post, while older material stays online after the underlying fact changes. Teams need checks before publication and a way to find affected content later, with evidence and human review.

02 Users affected by the problem

User group	Need
Brand / marketing	Consistent campaigns across teams and channels.
Creators / designers	Clear fixes that preserve the intent of their work.
Brand reviewers	Evidence, exception handling, and review history.
Product / communications	Update claims when facts or policies change.
Distributed teams	Shared, current guidance for employees and agencies.

03 Proposed solution

Dante links approved rules, facts, terms, and assets to the content that uses them. When a source changes, it surfaces connected items for review and shows recurring issues across review history.

Findings cite their source. Dante distinguishes an unsupported claim from one that conflicts with approved facts; reviewers make the final call and can record exceptions.

A demo scenario

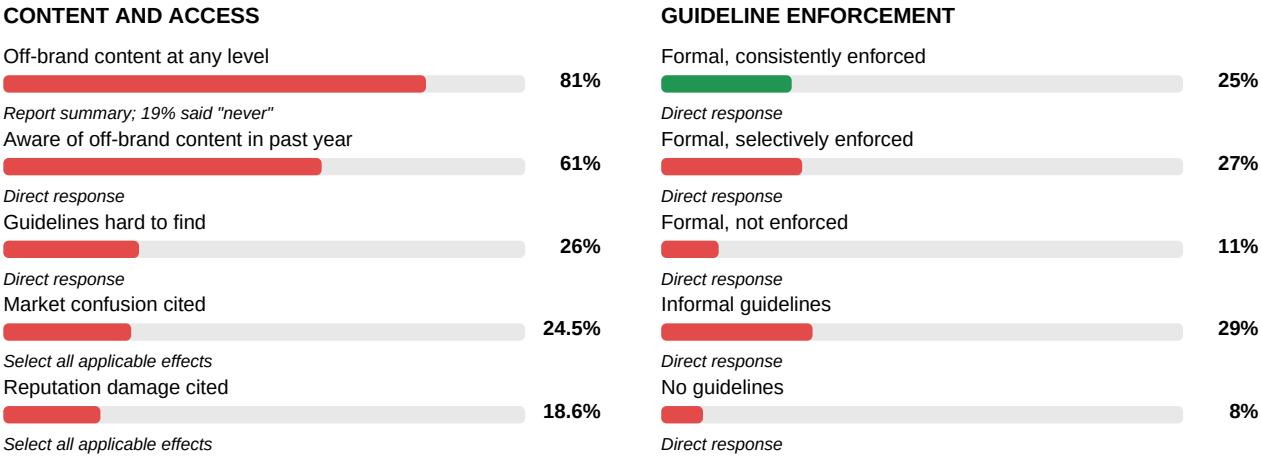
APPROVED FACT	FACT CHANGES	DANTE FLAGS FOR REVIEW
Pro plan storage: 100 GB	New approved value: 200 GB	Find linked drafts that still state 100 GB; show the source version and route each item to a reviewer.

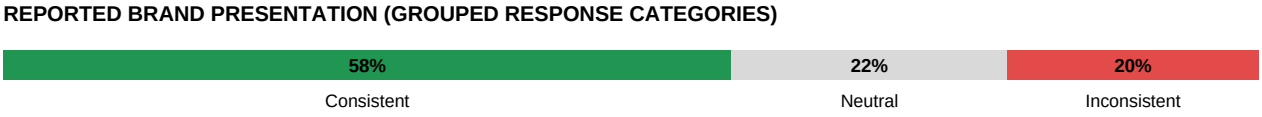
Published evidence

Lucidpress's 2019 survey is a historical, self-reported snapshot - useful context, not a current industry benchmark.

Published survey findings (Lucidpress, 2019)

Self-reported results. Left and right panels use different questions.





Respondents could select multiple effects. The figures describe perceptions, not measured financial losses.

Dante fills a practical gap: it connects a team's reviewed content to its approved sources, so a changed fact can lead to a targeted review.

Source: Lucidpress, *The State of Brand Consistency* (2019). Open report.

04 Complete product workflow

Stage	What happens
1. Add brand sources	Enter approved facts, rules, terms, and assets.
2. Approve versions	An administrator confirms each source and version.
3. Submit content	A creator submits text and selects its channel.
4. Review	Exact checks and source search find issues; AI helps explain them.
5. Resolve	Creators revise or request an exception; reviewers decide.
6. Keep history	Save findings, decisions, versions, and source links.
7. Recheck changes	When a source changes, list linked content for review.
8. Spot repeats	Group recurring findings and overrides over time.

05 Main features planned

Feature	Planned behaviour
Versioned sources	Approved facts, rules, terms, assets, and their history.
Evidence-linked review	Flagged text, source, explanation, and suggested fix.
Content links	See which approved sources each item uses.
Change impact	Find older content affected by a source update.
Human approvals	Comments, revisions, exceptions, and decisions.
Drift view	Recurring findings and overrides with examples.
Access controls	Workspace permissions and clear AI data handling.

06 Technology stack

Layer	Technology	Purpose
Web application	Next.js + TypeScript	Review interface, knowledge setup, impact view, and dashboard.
UI	Tailwind CSS + shadcn/ui	Consistent components and responsive screens.
Backend	Next.js Route Handlers	Authorization, review orchestration, versioning, and impact analysis.
Database / Auth / Storage	Supabase: PostgreSQL, Supabase Auth, private Storage	Store workspace data, user roles, review history, and private uploads.
AI	Gemini API or an available equivalent	Assist with rule extraction and semantic review; return structured, evidence-linked results.
Search	PostgreSQL full-text search	Retrieve relevant approved sources for the MVP; add vector search only if needed.
Charts / Hosting	Recharts; Vercel + Supabase	Visualise trends and deploy the demo.

Reliability approach

Use direct checks for exact terminology and structured facts. Use AI to help interpret wording, then verify that cited sources are approved and belong to the correct workspace. If the evidence is missing or unclear, send the item to a person.

07 Development plan across all three rounds

Round	Planned work and outcome
Round 1 - Proposal	Submit the concept, workflow, stack, and team plan.
Round 2 - Prototype	Build brand sources, text review, approvals, and one change-impact scenario.
Round 3 - Demo	Polish the 100 GB to 200 GB example; show affected content, evidence, reviewer action, and one measured drift pattern.

Align round tasks with the official schedule and judging criteria.

08 Round 2 - What we built

We turned the proposal into an interactive prototype with a complete source-to-review loop:

New in the prototype	What it does
Approved source library	Add guidance as pending, approve it, and retain the current version.
Evidence-linked checks	Check exact product facts and required terms; Gemini can assist with semantic rule review when enabled and explicitly selected.
Change impact preview	Preview which linked items still match or need attention before approving a changed source.
Human decisions and history	Revise content, record a reasoned exception, and keep source changes, findings, and decisions in an audit trail.
Persistent workspace and analytics	Save authenticated workspace state in Supabase Postgres and show review activity with Recharts.

The demo uses linked sample content to make the workflow repeatable. It does not automatically crawl live websites or publishing systems.

09 Round 3 - Expected

Next, we would move from a strong guided prototype toward a workflow teams can connect to real channels:

Potential next feature	Round 3 direction
Connect real content sources	Add website/CMS connectors and scheduled checks, with clear permissions and a preview before any action.
Bring in team guidance	Import PDF/DOCX policies with page-level evidence and reviewer approval before they become trusted sources.
Support team review	Add shared workspaces, reviewer assignments, approval roles, and notifications.
Measure quality over time	Add channel-level drift trends and evaluate checks against a reviewed test set; report measured precision and misses.
Broaden coverage	Explore multilingual rules and image/logo checks, with uncertain results routed to a person.

These are proposed extensions, not features currently claimed as working. We would prioritise integrations and document evidence first, then expand after testing reviewer accuracy.

10 Expected impact

Dante aims to catch stale or unsupported content earlier and cut time spent searching for current guidance. The prototype demonstrates how a changed fact can lead to a targeted review; impact claims should be measured with pilot users before reporting.

11 Future scope

- Ingest PDF and DOCX guidelines with administrator approval.
- Check images for text, logo versions, and contrast.
- Track drift by campaign, channel, and team over time.
- Connect to publishing tools with configurable approval gates.
- Add multilingual rules, audit exports, and finer access controls.

12 Team details

Chrollo | CodeNexus | NSEC
Team contact: Samriddha Chaudhury - samriddhachaudhury@gmail.com

Member	**Role**
Samriddha Chaudhury	Team Lead; Full Stack Developer
Md Rizwaan Rahaman	Full Stack Developer

13 Planned contributions

Samriddha Chaudhury, Team Lead: product scope, backend, data model, source links, and change-impact logic.

Md Rizwaan Rahaman: frontend screens, review workflow, dashboard, and demo content. Both will integrate, validate, and present the prototype.